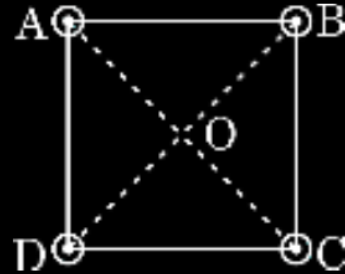


3. Four long straight thin wires are held vertically at the corners A, B, C and D of a square of side 'a', kept on a table and carry equal current 'I'. The wire at A carries current in upward direction whereas the current in the remaining wires flows in downward direction. The net magnetic field at the centre of the square will have the magnitude :

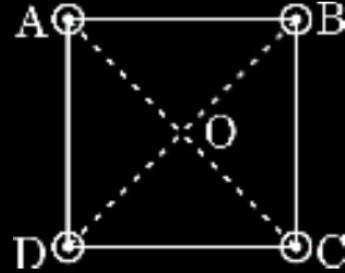
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- (A) $\frac{\mu_0 I}{\pi a}$ and directed along OC
- (B) $\frac{\mu_0 I}{\pi a \sqrt{2}}$ and directed along OD
- (C) $\frac{\mu_0 I \sqrt{2}}{\pi a}$ and directed along OB
- (D) $\frac{2 \mu_0 I}{\pi a}$ and directed along OA

3. Four long straight thin wires are held vertically at the corners A, B, C and D of a square of side 'a', kept on a table and carry equal current 'I'. The wire at A carries current in upward direction whereas the current in the remaining wires flows in downward direction. The net magnetic field at the centre of the square will have the magnitude :

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- (A) $\frac{\mu_0 I}{\pi a}$ and directed along OC
- (B) $\frac{\mu_0 I}{\pi a \sqrt{2}}$ and directed along OD
- (C) $\frac{\mu_0 I \sqrt{2}}{\pi a}$ and directed along OB
- (D) $\frac{2 \mu_0 I}{\pi a}$ and directed along OA

(C) $\frac{\mu_0 I}{\pi a} \sqrt{2}$ and directed along OB

23. (a) A straight vertical wire carries a current of 3 A upward along +z axis. Calculate the magnitude and direction of the magnetic field at a point 15 cm away from the wire on +x-axis. Name the rule used to find the direction of magnetic field.

23. (a) A straight vertical wire carries a current of 3 A upward along +z axis. Calculate the magnitude and direction of the magnetic field at a point 15 cm away from the wire on +x-axis. Name the rule used to find the direction of magnetic field.

3

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$$\begin{aligned}
 \text{(a)} \quad B &= \frac{\mu_0}{4\pi} \frac{2I}{r} \\
 &= \frac{(10^{-7})(2)(3)}{15 \times 10^{-2}} \\
 &= 4 \times 10^{-6} \text{ T}
 \end{aligned}$$

Direction: Along Y-axis or $\hat{j}$

Rule: Right hand thumb rule or Maxwell screw rule

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

5. A straight long wire lying along y -axis, carries a current of 1 A along $-y$ direction. The magnetic field due to the conductor at a point (50 cm, 0, 0) will point along.

(A) z -axis

(B) $-z$ -axis

(C) x -axis

(D) $-x$ -axis

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5. A straight long wire lying along y -axis, carries a current of 1 A along $-y$ direction. The magnetic field due to the conductor at a point (50 cm, 0, 0) will point along.

(A) z -axis
(C) x -axis

(B) $-z$ -axis
(D) $-x$ -axis

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1

(A) z -axis

5. A person is standing facing geographic north and a vertical wire in front of him is carrying a current in downward direction. The direction of magnetic field at a point on the right side of wire is

1

(A) South

(B) East

(C) West

(D) North

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5. A person is standing facing geographic north and a vertical wire in front of him is carrying a current in downward direction. The direction of magnetic field at a point on the right side of wire is

1

(A) South

(B) East

(C) West

(D) North

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(A) South

3. A long straight wire of circular cross-section (radius a) carries a steady current I . The current is uniformly distributed across this cross-section. The magnitude of the magnetic field produced at a point at a distance $(a/2)$ from the axis of the wire will be

1

- (A) Zero
- (B) $\frac{\mu_0 I}{2\pi a}$
- (C) $\frac{\mu_0 I}{4\pi a}$
- (D) $\frac{\mu_0 I}{6\pi a}$

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3. A long straight wire of circular cross-section (radius a) carries a steady current I . The current is uniformly distributed across this cross-section. The magnitude of the magnetic field produced at a point at a distance $(a/2)$ from the axis of the wire will be

1

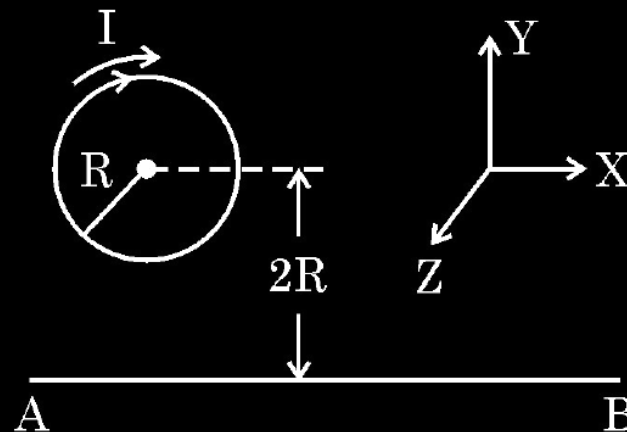
- (A) Zero
- (B) $\frac{\mu_0 I}{2\pi a}$
- (C) $\frac{\mu_0 I}{4\pi a}$
- (D) $\frac{\mu_0 I}{6\pi a}$

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(C) $\frac{\mu_0 I}{4\pi a}$

10. A circular loop has radius R and carries current I as shown in figure. In order that the net magnetic field at the centre of the loop is zero, the current in wire AB should have magnitude

1

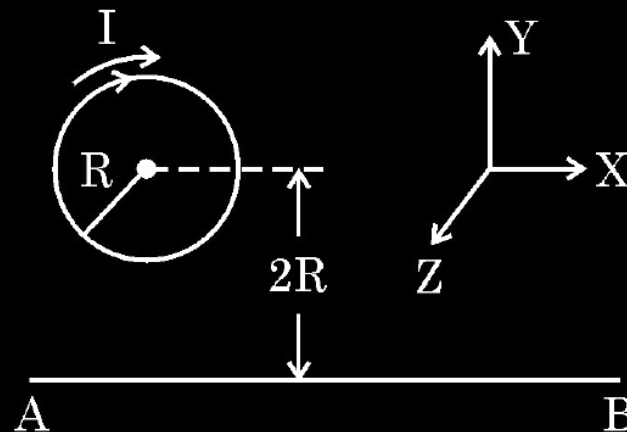


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- (A) $2\pi I$, along $+X$ – axis
- (B) $2\pi I$ along $-X$ – axis
- (C) πI along $+X$ – axis
- (D) πI along $-X$ – axis

10. A circular loop has radius R and carries current I as shown in figure. In order that the net magnetic field at the centre of the loop is zero, the current in wire AB should have magnitude

1



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- (A) $2\pi I$, along $+X$ – axis
- (B) $2\pi I$ along $-X$ – axis
- (C) πI along $+X$ – axis
- (D) πI along $-X$ – axis

(ii) A long straight wire kept horizontally carries a current of 4 A from west to east direction. Find the direction and magnitude of magnetic field $\vec{B}$ at a point 20 cm below the wire.

29. Two infinitely long parallel conductors A and B, separated by 2 cm, carry 6 A and 2 A currents in opposite directions respectively. Find :

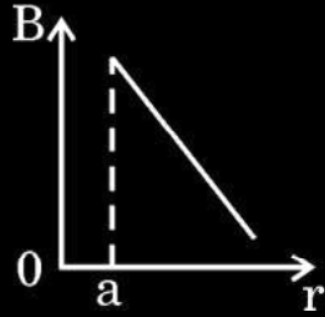
- (a) the net magnetic field at a point midway between A and B.
- (b) the force acting per unit length on B.

3

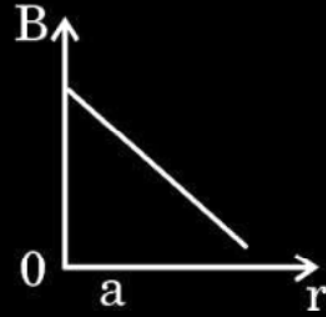
(a) Calculation of magnetic field due to current in A and B	$\frac{1}{2} + \frac{1}{2}$
Net magnetic field	1
(b) Force per unit length	1

(a) $B_1 = \frac{\mu_0 I_1}{2\pi r} = \frac{2 \times 10^{-7} \times 6}{1 \times 10^{-2}}$ $= 1.2 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2}$
$B_2 = \frac{\mu_0 I_2}{2\pi r} = \frac{2 \times 10^{-7} \times 2}{1 \times 10^{-2}}$ $= 0.4 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2}$
$\therefore B_1$ and B_2 are in same direction. $B_{net} = B_1 + B_2$ $= 1.2 \times 10^{-4} \text{ T} + 0.4 \times 10^{-4} \text{ T}$ $= 1.6 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2} + \frac{1}{2}$
(b) Magnitude of magnetic force per unit length	
$\frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$	$\frac{1}{2}$
$= \frac{2 \times 10^{-7} \times 6 \times 2}{2 \times 10^{-2}} = 1.2 \times 10^{-4} \text{ N/m}$	$\frac{1}{2}$

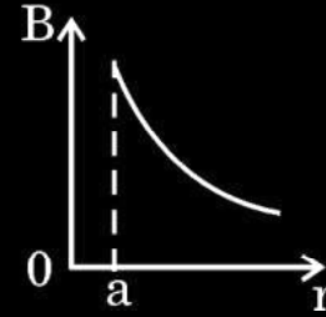
15. Which of the following graphs correctly represents the variation of the magnitude of the magnetic field outside a straight infinite current carrying wire of radius 'a', as a function of distance 'r' from the centre of the wire ?



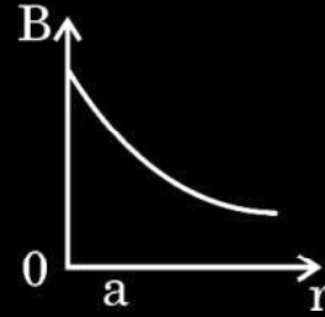
(a)



(b)

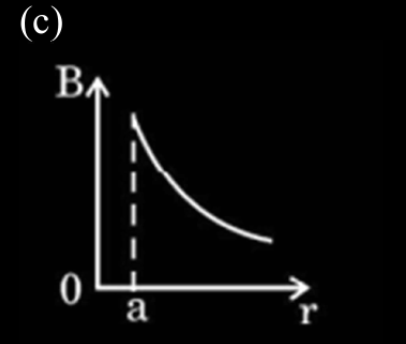
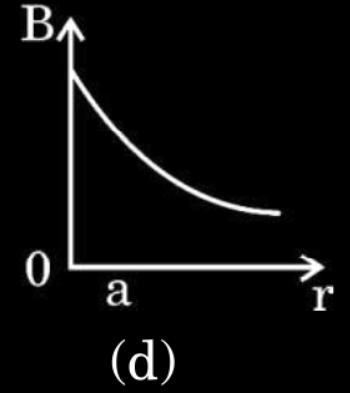
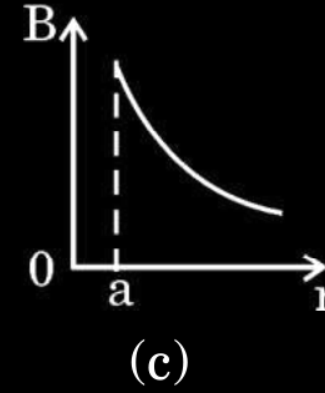
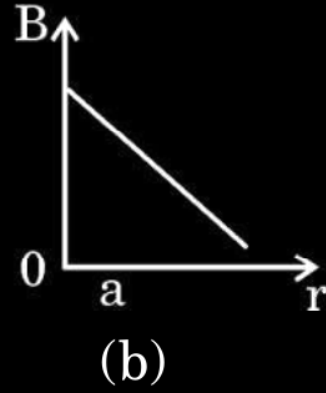
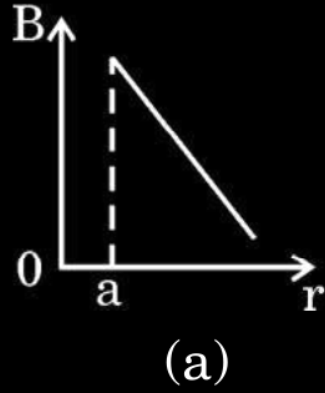


(c)



(d)

15. Which of the following graphs correctly represents the variation of the magnitude of the magnetic field outside a straight infinite current carrying wire of radius 'a', as a function of distance 'r' from the centre of the wire ?



1. A long straight wire of radius 'a' carries a steady current 'I'. The current is uniformly distributed across its area of cross-section. The ratio of magnitude of magnetic field $\vec{B}_1$ at $\frac{a}{2}$ and $\vec{B}_2$ at distance 2a is

1

(a) $\frac{1}{2}$

(b) 1

(c) 2

(d) 4

1. A long straight wire of radius 'a' carries a steady current 'I'. The current is uniformly distributed across its area of cross-section. The ratio of magnitude of magnetic field $\vec{B}_1$ at $\frac{a}{2}$ and $\vec{B}_2$ at distance 2a is **1**

(a) $\frac{1}{2}$

(b) 1

(c) 2

(d) 4

2. Two horizontal thin long parallel wires, separated by a distance r carry current I each in the opposite directions. The net magnetic field at a point midway between them, will be

1

- (A) zero
- (B) $\left(\frac{\mu_0 I}{2\pi r}\right)$ vertically downward
- (C) $\left(\frac{2\mu_0 I}{r}\right)$ vertically upward
- (D) $\left(\frac{\mu_0 I}{\pi r}\right)$ vertically downward

2. Two horizontal thin long parallel wires, separated by a distance r carry current I each in the opposite directions. The net magnetic field at a point midway between them, will be

1

- (A) zero
- (B) $\left(\frac{\mu_0 I}{2\pi r}\right)$ vertically downward
- (C) $\left(\frac{2\mu_0 I}{r}\right)$ vertically upward
- (D) $\left(\frac{\mu_0 I}{\pi r}\right)$ vertically downward

No option is correct.

[Award one mark to each student]

- (ii) A long horizontal conductor carries a current of 30 A, in east to west direction. What are the magnitude and direction of the magnetic field due to the current 1.0 m below the conductor ?

(ii) $B = \frac{2\mu_o I}{4\pi r}$	$\frac{1}{2}$
$B = \frac{2 \times 10^{-7} \times 30}{1} = 0.6 \times 10^{-5} T$	$\frac{1}{2}$
Magnetic field is directed towards SOUTH	1

17. *Assertion (A) :* The magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason (R): The magnetic field inside the solenoid is uniform.

17. *Assertion (A)* : The magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason (R): The magnetic field inside the solenoid is uniform.

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

22. (b) (i) State and explain Ampere's circuital law.
- (ii) Two long straight parallel wires separated by 20 cm, carry 5 A and 10 A current respectively, in the same direction. Find the magnitude and direction of the net magnetic field at a point midway between them.

(i) Statement and explanation of Ampere's circuital law	1
(ii) Finding magnitude and direction of magnetic field	2

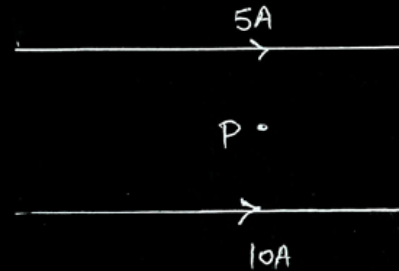
Line integral of magnetic field over a closed loop in vacuum is equal to μ_0 times the total current passing through the loop.

Alternatively

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

The integral in this expression is over a closed loop coinciding with the boundary of the surface.

(ii)



$$B = \frac{\mu_0 I}{2\pi r}$$

Net magnetic field $B = B_2 - B_1$

$$B = \frac{\mu_0 \times 10^2}{20\pi} [10 - 5]$$

$$B = \frac{4\pi \times 10^{-7} \times 10^2 \times 5}{20\pi}$$

$$B = 10^{-5} T$$

Along the direction of magnetic field produced by the conductor carrying current 10A.

1

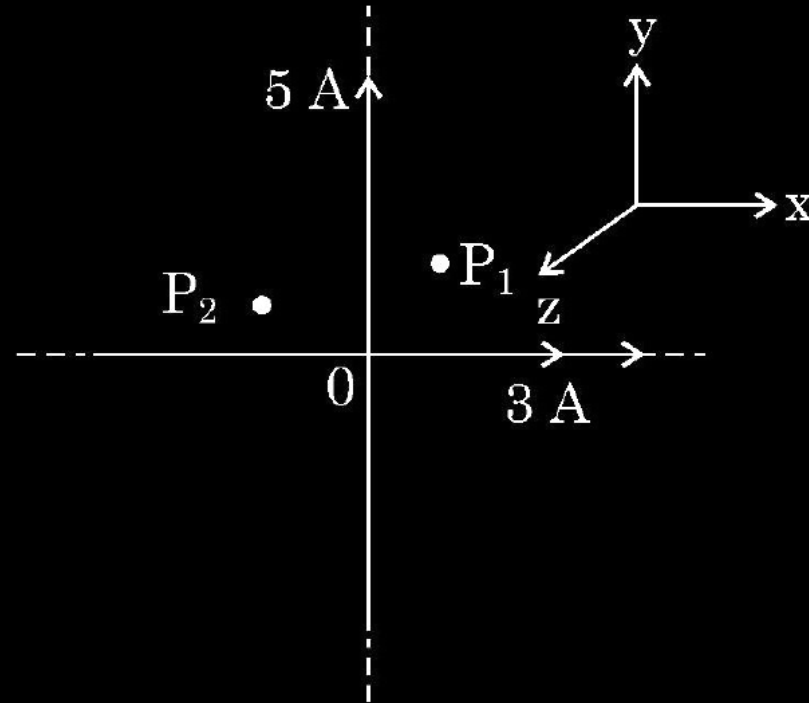
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

25. Two long insulated straight wires carrying currents of 3 A and 5 A are arranged in XY plane as shown in figure. Find the magnitude and direction of the net magnetic fields at points $P_1(2\text{m}, 2\text{m})$ and $P_2(-1\text{m}, 1\text{m})$. 3



Net magnetic field at point P₁

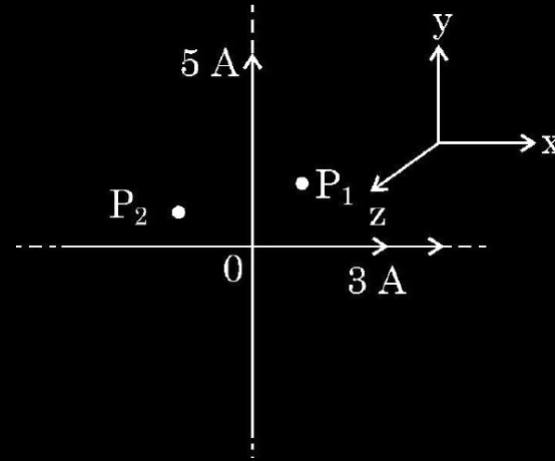
$$\begin{aligned}
 B &= B_{y(\text{wire})} - B_{x(\text{wire})} \\
 &= \frac{\mu_0 I_1}{2\pi r} - \frac{\mu_0 I_2}{2\pi r} \\
 &= \frac{\mu_0}{2\pi \times 2} (5 - 3) \\
 B &= 2 \times 10^{-7} \text{ T}
 \end{aligned}$$

The direction of net magnetic field is along -ve z-axis.

Net magnetic field at point P₂

$$\begin{aligned}
 B &= B_{y(\text{wire})} + B_{x(\text{wire})} \\
 &= \frac{\mu_0 I_1}{2\pi r} + \frac{\mu_0 I_2}{2\pi r} \\
 &= \frac{4\mu_0}{2\pi \times 1} (5 + 3) \\
 &= \frac{4\mu_0}{\pi} \\
 &= 16 \times 10^{-7} \text{ T}
 \end{aligned}$$

The direction of net magnetic field is along +ve z-axis.



1/2

1/2

1/2

1/2

1/2

1/2

5. A straight wire is kept horizontally along east-west direction. If a steady current flows in wire from east to west, the magnetic field at a point above the wire will point towards

(A) East

(B) West

(C) North

(D) South

5. A straight wire is kept horizontally along east-west direction. If a steady current flows in wire from east to west, the magnetic field at a point above the wire will point towards

(A) East

(B) West

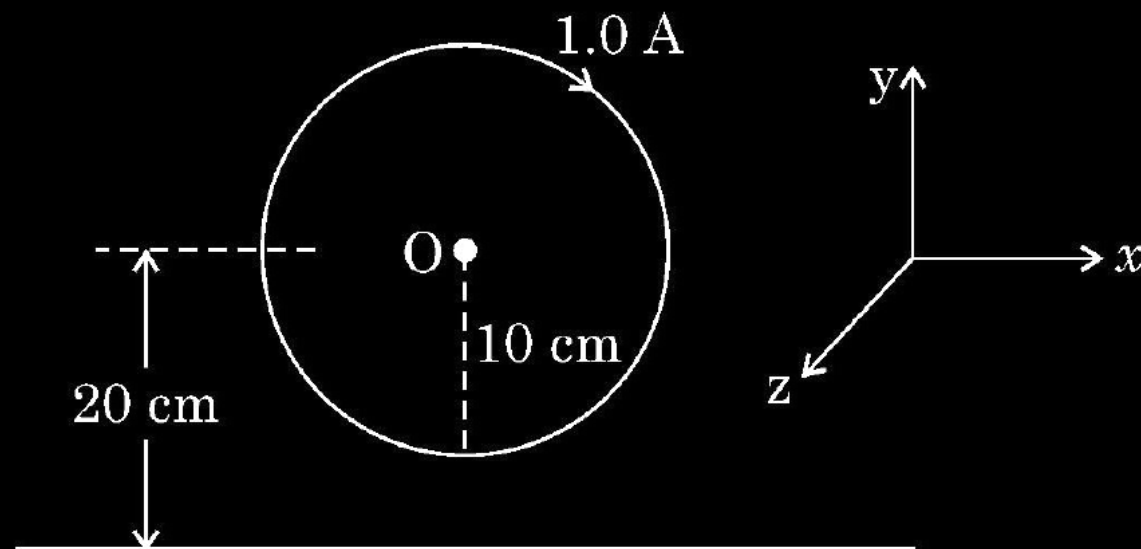
(C) North

(D) South

1

(C) North

26. A circular loop of radius 10 cm carrying current of 1.0 A lies in x - y plane. A long straight wire lies in the same plane parallel to x -axis at a distance of 20 cm as shown in figure.



Find the direction and value of current that has to be maintained in the wire so that the net magnetic field at O is zero.

Magnetic field due to circular current loop at its centre O.

$$B_1 = \frac{\mu_0 I_1}{2r}$$

$$= \frac{\mu_0 \times 1}{2 \times 0.1}$$

$$= 5\mu_0 \text{ T}$$

The magnetic field B_1 is perpendicular to plane of loop and directed inwards.

Magnetic field due to long current carrying straight wire at O.

$$B_2 = \frac{\mu_0 I_2}{2\pi r}$$

$$B_2 = \frac{\mu_0 I_2}{2\pi \times 0.2} \text{ T}$$

For net magnetic field at O to be zero, B_1 should be equal and opposite to B_2 .

$$5\mu_0 = \frac{\mu_0 I_2}{0.4\pi}$$

$$I_2 = 2\pi \text{ A}$$

$$= 6.28 \text{ A}$$

Direction of current in the straight wire is along +ve x axis.

Alternatively

Net magnetic field at O is zero.

$$B_{\text{loop}} = B_{\text{wire}}$$

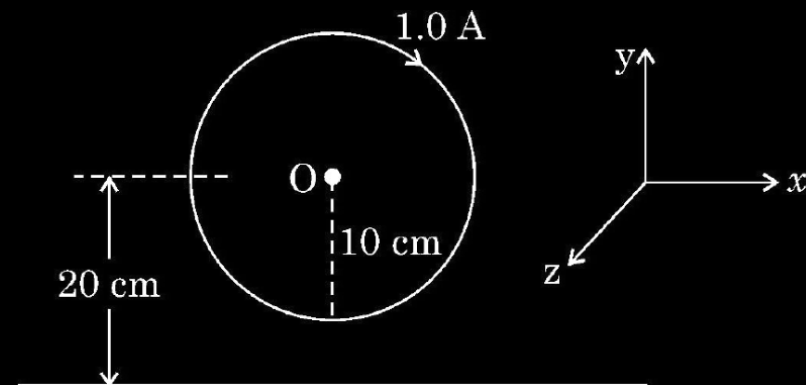
$$\frac{\mu_0 I_1}{2r_1} = \frac{\mu_0 I_2}{2\pi r_2}$$

$$\frac{\mu_0 \times 1}{2 \times 0.1} = \frac{\mu_0 I_2}{2\pi \times 0.2}$$

$$I_2 = 2\pi \text{ A}$$

$$= 6.28 \text{ A}$$

Direction of current is along + x-axis.



4. The magnetic field B within the solenoid of length L having N turns and carrying current I will be :

(A) $\mu_0 N I L$	(B) $\frac{\mu_0 N I}{L}$
(C) $\frac{\mu_0 N I}{4\pi L}$	(D) $\mu_0 N I$

4. The magnetic field B within the solenoid of length L having N turns and carrying current I will be :

(A) $\mu_0 N I L$ (B) $\frac{\mu_0 N I}{L}$

(C) $\frac{\mu_0 N I}{4\pi L}$ (D) $\mu_0 N I$

(B) $\frac{\mu_0 N I}{L}$

32. (a) (A) When a current passes through a conductor it produces a magnetic field.
- (i) State the right-hand thumb rule that gives the direction of the magnetic field.
 - (ii) How does the magnetic field vary with the distance from the conductor ?
 - (iii) Name the factors (other than the distance) which affect the magnitude of the magnetic field.

(A) (i) Stating the right-hand thumb rule	1
(ii) Variation of magnetic field with distance	1
(iii) Naming the factors affecting magnitude of magnetic field	1

(a) (i) Right hand thumb rule

When we hold a conductor in right hand in such a way that a thumb stretches in the direction of flow of current then the curled fingers show the direction of magnetic field.

1

(ii) Magnetic field due to straight conductor at a distance r is given by

Since $B \propto \frac{1}{r}$ As r increases, B decreases

1

(iii) B depends on current, nature of medium and angle θ between current element and displacement r . (any two factors)

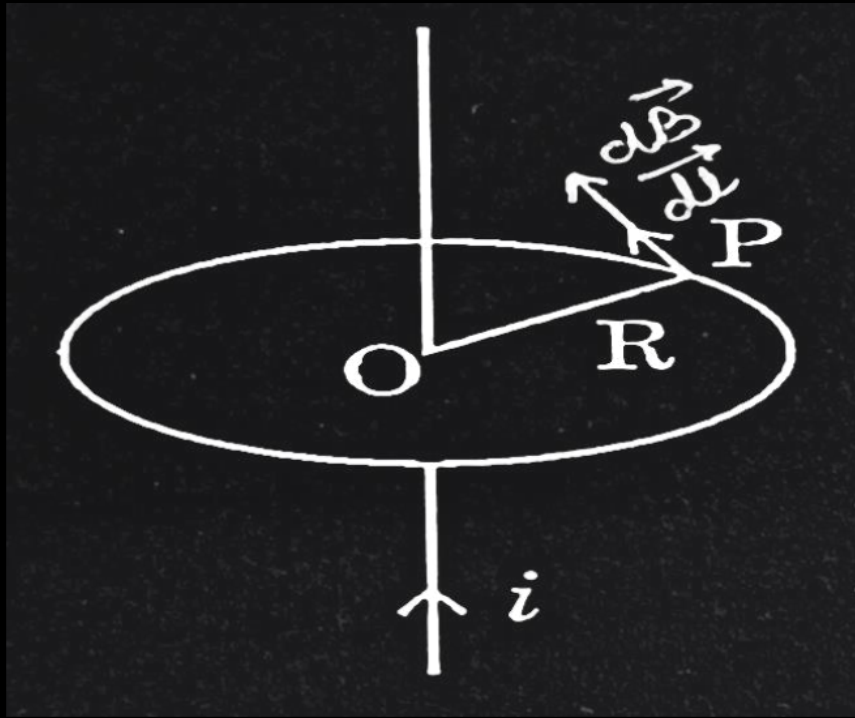
$\frac{1}{2} + \frac{1}{2}$

- 27.** (a) Use Ampere's law to derive the expression for the magnetic field due to a long straight current carrying wire of infinite length.
- (b) Why is Ampere's law used for the derivation in (a) above and not Biot-Savart's law ? Explain.

3

(a) Deriving expression of magnetic field for long straight wire	2
(b) Explanation	1

(a)



$\frac{1}{2}$

Consider an amperian loop of radius R around the wire carrying current I .

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$\frac{1}{2}$

$$B \times 2\pi r = \mu_0 i$$

$\frac{1}{2}$

$$B = \frac{\mu_0 i}{2\pi r}$$

$\frac{1}{2}$

(b) The expression for magnetic field due to a long straight current carrying infinite wire can be obtained easily using Ampere's circuital law whereas Biot- Savart's law requires lengthy calculation.

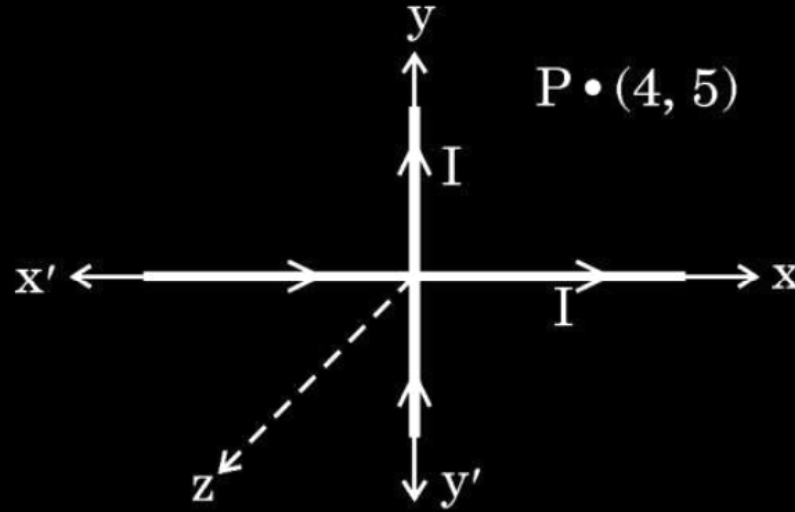
1

8. A long straight wire is held vertically and carries a steady current in upward direction. The shape of magnetic field lines produced by the current-carrying wire are :
- (A) horizontal straight lines directed radially out from the wire.
 - (B) straight lines parallel to the current-carrying wire.
 - (C) concentric horizontal circles around the wire.
 - (D) coaxial helixes around the wire.

8. A long straight wire is held vertically and carries a steady current in upward direction. The shape of magnetic field lines produced by the current-carrying wire are :
- (A) horizontal straight lines directed radially out from the wire.
 - (B) straight lines parallel to the current-carrying wire.
 - (C) concentric horizontal circles around the wire.
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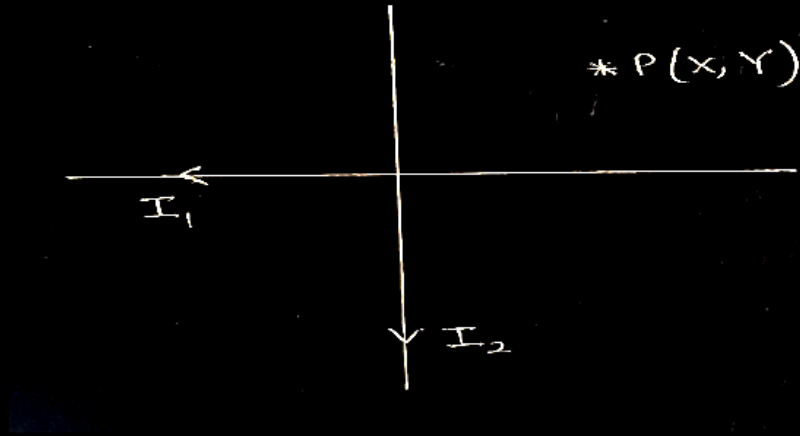
(C) Concentric horizontal circles around the wire

23. (a) Write vector form of Biot-Savart law.
- (b) Two insulated long straight wires, each carrying 2.0 A current are kept along xx' and yy' axis as shown in the figure. Find the magnitude and direction of resultant magnetic field at point P (4m, 5m).



(a) $d\vec{B} = \frac{\mu_o}{4\pi} \frac{I(d\vec{l} \times \vec{r})}{r^3}$	1
(b) As $B = \frac{\mu_o}{4\pi} \frac{2I}{a}$	
$\therefore B_1 = \text{Magnetic field due to current carrying wire along } XX'$	
$\vec{B}_1 = \frac{\mu_o}{4\pi} \frac{2I}{a} = \frac{2 \times 2 \times 10^{-7}}{5} = (8 \times 10^{-8} \text{ T})(\hat{k})$	$\frac{1}{2}$
$B_2 = \text{Magnetic field due to current carrying wire along } YY'$	
$\vec{B}_2 = \frac{\mu_o}{4\pi} \frac{2I}{a} = \frac{2 \times 2 \times 10^{-7}}{4} = 10 \times 10^{-8} \text{ T } (-\hat{k})$	$\frac{1}{2}$
$\vec{B}_{\text{net}} = \vec{B}_2 + \vec{B}_1$	$\frac{1}{2}$
$= 2 \times 10^{-8} \text{ T } (-\hat{k})$	
• Direction of resultant magnetic field along $-Z$ axis.	$\frac{1}{2}$

- (b) Two infinitely long conductors kept along XX' and YY' axes are carrying current I_1 and I_2 along $-X$ axis and $-Y$ axis respectively. Find the magnitude and direction of the net magnetic field produced at point $P(X, Y)$.



Magnetic field due to conductor carrying current I_1 ($\vec{B}_1$) = $\frac{\mu_0 I_1}{2\pi Y} (-\hat{k})$

1/2

Magnetic field due to conductor Carrying current I_2 ($\vec{B}_2$) = $\frac{\mu_0 I_2}{2\pi X} (\hat{k})$

1/2

$$\vec{B}_p = \vec{B}_1 + \vec{B}_2$$

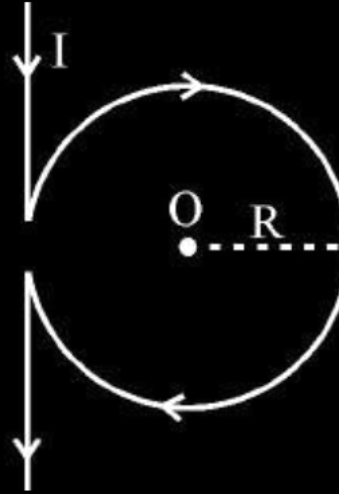
$$\vec{B}_p = \frac{\mu_0}{2\pi} \left[\frac{I_2}{X} - \frac{I_1}{Y} \right] \hat{k}$$

1

Direction will be along the Z-axis.

1

3. A current I flows through a long straight conductor which is bent into a circular loop of radius R in the middle as shown in the figure.



The magnitude of the net magnetic field at point O will be

- (a) Zero (b) $\frac{\mu_0 I}{2R} (1 + \pi)$ (c) $\frac{\mu_0 I}{4\pi R}$ (d) $\frac{\mu_0 I}{2R} \left(1 - \frac{1}{\pi}\right)$ **1**

3. A current I flows through a long straight conductor which is bent into a circular loop of radius R in the middle as shown in the figure.



The magnitude of the net magnetic field at point O will be

- (a) Zero (b) $\frac{\mu_0 I}{2R} (1 + \pi)$ (c) $\frac{\mu_0 I}{4\pi R}$ (d) $\frac{\mu_0 I}{2R} \left(1 - \frac{1}{\pi}\right)$ **1**

(d) $\frac{\mu_0 I}{2R} \times \left(1 - \frac{1}{\pi}\right)$

7. A current of 10A is flowing from east to west in a long straight wire kept on a horizontal table. The magnetic field developed at a distance of 10 cm due north on the table is :

- (a) 2×10^{-5} T, acting downwards (b) 2×10^{-5} T, acting upwards
(c) 4×10^{-5} T, acting downwards (d) 4×10^{-5} T, acting upwards

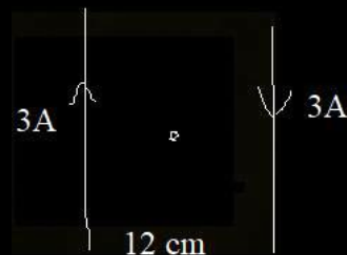
7. A current of 10A is flowing from east to west in a long straight wire kept on a horizontal table. The magnetic field developed at a distance of 10 cm due north on the table is :

- (a) 2×10^{-5} T, acting downwards (b) 2×10^{-5} T, acting upwards
(c) 4×10^{-5} T, acting downwards (d) 4×10^{-5} T, acting upwards

1

(a) 2×10^{-5} T acting downwards

- (b) Two long parallel straight conductors are placed 12 cm apart in air. They carry equal currents of 3 A each. Find the magnitude and direction of the magnetic field at a point midway between them (drawing a figure) when the currents in them flow in opposite directions.



$$\vec{B} = \vec{B}_1 + \vec{B}_2$$

$$B = \frac{\mu_o I_1}{2\pi r_1} + \frac{\mu_o I_2}{2\pi r_2}$$

$$= \frac{\mu_o}{2\pi} \left(\frac{3}{6 \times 10^{-2}} + \frac{3}{6 \times 10^{-2}} \right)$$

$$= \frac{4\pi \times 10^{-7} \times 3}{\pi \times 6 \times 10^{-2}}$$

$$= 2 \times 10^{-5} \text{ tesla}$$

Direction of $\vec{B}$ at midpoint is perpendicular to the plane containing the two conductors and pointing downwards.

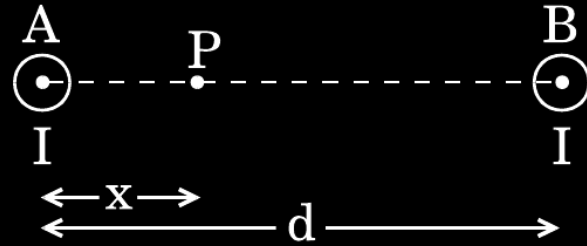
(Note: give full credit of this direction if student takes direction opposite to the shown in fig and answer accordingly)

1/2

1

1/2

25. Two long straight parallel wires A and B separated by a distance d , carry equal current I flowing in same direction as shown in the figure.



- (a) Find the magnetic field at a point P situated between them at a distance x from one wire.
- (b) Show graphically the variation of the magnetic field with distance x for $0 < x < d$.

a)

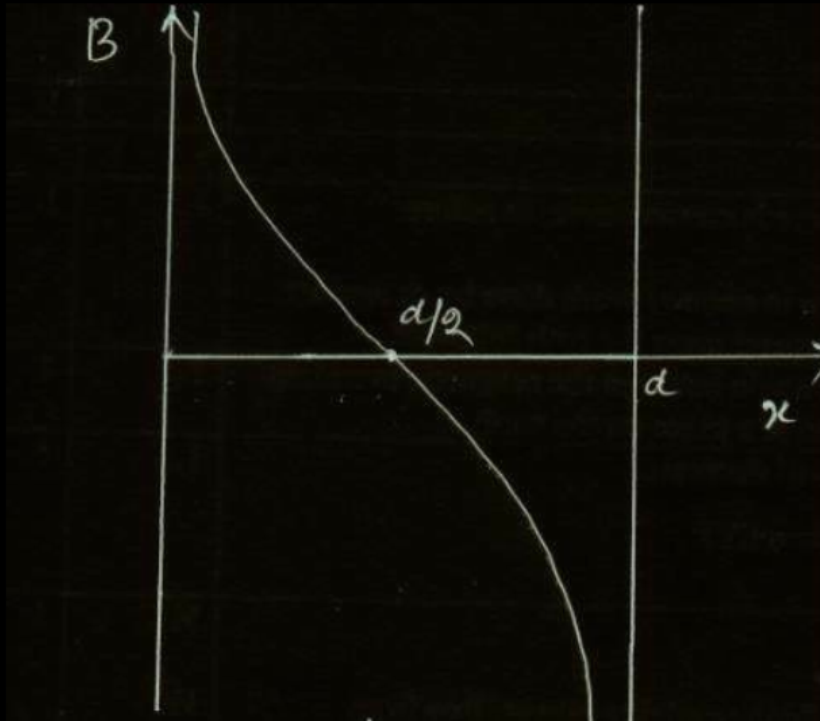
$$B = \frac{\mu_0 I}{2\pi x}$$

$$B_P = B_1 - B_2 = \frac{\mu_0 I}{2\pi x} - \frac{\mu_0 I}{2\pi(d-x)} = \frac{\mu_0 I(d-2x)}{2\pi(d-x)x}$$

$\frac{1}{2}$

1

b)



$\frac{1}{2}$

State Ampere's circuital law. Use this law to find magnetic field due to straight infinite current carrying wire. How are the magnetic field lines different from the electrostatic field lines ?

Statement of Ampere's Circuital law	1
Finding Magnetic Field	1 ½
Differences between the two types of field lines	½

According to Ampers's circuital law, " The line integral of the magnetic field, around a closed loop, equals μ_o times the total current passing through the surface enclosed by that loop."

1

Alternatively,

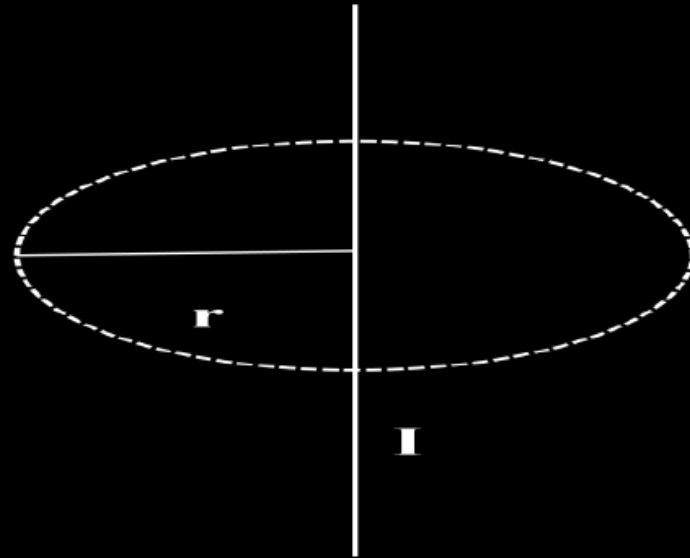
$$\oint \vec{B} \cdot d\vec{l} = \mu_o I$$

For the infinite current carrying wire, we get

$$B \oint dl = \mu_o I$$

$$\text{or } B 2\pi r = \mu_o I$$

$$\text{or } B = \frac{\mu_o I}{2\pi r}$$



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

The magnetic field lines form closed loops while the electric field lines originate from positive charges and end at negative charges.

$\frac{1}{2}$